

EMANUELE PALUMBO

PhD Student in Machine Learning @ ETH Zurich | [epalu.github.io](https://github.com/epalu)

Wehntalerstrasse 30, 8057 Zurich, CH | +41 76 530 3180 | Birthdate: 17.09.1996 | Nationality: Italian

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EXPERIENCE

Apple AIML

April 2025 – November 2025

ML Research Scientist Intern - Health AI

Zurich, Switzerland

- Designed a novel generative model trained on multimodal biosignal data to generate arterial pressure waveform signals from non-invasive photoplethysmography measurements, for non-invasive assessment of cardiac function
- Demonstrated inference of key biomarkers from generated signals using a predictive model trained solely on hemodynamic simulations, enabling cardiovascular biomarker estimation without the need for labeled data with invasive annotations
- Research project selected for presentation at Apple Park Cupertino, underscoring both technical impact and relevance in the company
- Paper with research outcomes currently under review
- Contributed to a library to scale experiments on internal data. Implemented key features that enhanced efficiency in launching and reproducing experiments across the team.

EDUCATION

ETH Zurich

2022 – present

PhD Candidate, Institute for Machine Learning, Dept. of Computer Science

Zurich, Switzerland

- Doctoral Fellow of the ETH AI Center
- Member of the Medical Data Science Group led by Prof. Julia Vogt
- **Key research areas:** Multimodal Learning, Generative Models, Unsupervised Learning, AI for Health

ETH Zurich

2018 – 2021

M. Sc. Data Science

Zurich, Switzerland

- GPA: 5.49/6
- Master Thesis: *Structured Mixture-of-experts Multimodal Variational Autoencoders*

Università Politecnica delle Marche

2015 – 2018

B. Sc. Computer and Automation Engineering

Ancona, Italy

- Final Grade : 110/110 *cum laude*
- Bachelor Thesis: *A MILP model for the maximum connected quasi-clique problem*

PUBLICATIONS [†]

- Deep Generative Clustering with Multimodal Diffusion Variational Autoencoders ([↗](#))
Palumbo, E., Manduchi, L., Laguna, S., Chopard, D., and Vogt, J. E.
International Conference on Learning Representations, 2024
- MMVAE+: Enhancing the Generative Quality of Multimodal VAEs without Compromises ([↗](#))
Palumbo, E., Daunhawer, I., and Vogt J. E.
International Conference on Learning Representations, 2023

- Effective Bayesian Heteroscedastic Regression with Deep Neural Networks ([↗](#))
Immer, A.* , **Palumbo, E.***, Marx A., and Vogt, J. E.
Advances in Neural Information Processing Systems, 2023
- From Logits to Hierarchies: Hierarchical Clustering made Simple ([↗](#))
Palumbo, E., Vandenhirtz M., Ryser A., Daunhawer I., and Vogt, J. E.
International Conference on Machine Learning, 2025
- Identifiability Results for Multimodal Contrastive Learning ([↗](#))
Daunhawer, I., Bizeul, A., **Palumbo, E.**, Marx, A., and Vogt, J. E.
International Conference on Learning Representations, 2023
- On the Limitations of Multimodal VAEs ([↗](#))
Daunhawer I., Sutter T. M., Chin-Cheong, K., **Palumbo, E.**, and Vogt, J. E.
International Conference on Learning Representations, 2022
- 3DIdentBox: A Toolbox for Identifiability Benchmarking. ([↗](#))
Bizeul, A., Daunhawer, I., **Palumbo, E.**, Schölkopf, B., Marx, A., and Vogt, J. E.
CleaR (Datasets Track) 2023.
- Therapeutic stays of Belarusian children in Italy: evaluation of their mental status, psychological consequences and physical health status. ([↗](#))
Ferrara, P., Pianese, G., Franceschini, G., **Palumbo, E.**, Ianni, A., Ghilardi, G..
Minerva Pediatrics, 2021

* Equal contribution

† Preprints and Workshop Publications are listed on Google Scholar ([↗](#))

AWARDS AND ACHIEVEMENTS

- Nov. 2024 **Top Reviewer**
NeurIPS 2024 ([↗](#))
- Dec. 2023 **Organizer and Program Chair**
Deep Generative Models for Health workshop @ NeurIPS 2023 ([↗](#))
- Oct. 2023 **Top Reviewer**
NeurIPS 2023 ([↗](#))
- May 2023 **Organizer and Program Chair**
Time Series Representation Learning for Health workshop @ ICLR 2023 ([↗](#))
- May 2022 **ETH AI Center Doctoral Fellowship**
Recipient of the highly competitive ETH AI Center Doctoral Fellowship ([↗](#))
Supervised by Prof. Julia Vogt, with the co-supervision of Prof. Andrea Burden

RECENT TALKS

- 12/03/2024 **Invited talk at IBM Research Zurich, AI for Scientific Discovery**
Generative Models for Representation Learning with Multiple Heterogeneous Modalities

REVIEWING EXPERIENCE

I reviewed for NeurIPS, ICML, and ICLR conferences, and was recognized as a Top Reviewer for NeurIPS 2023 and NeurIPS 2024.

TECHNICAL SKILLS

Programming Languages	Proficient with Python . Experienced in Bash . Familiar with R , Matlab , C++ , SQL .
Deep Learning	Proficient with PyTorch , TensorFlow , HuggingFace .
Machine Learning	Proficient with NumPy , Pandas , Scikit-learn , Matplotlib . Familiar with OpenCV , SciPy , NLTK .
Software Engineering & Other	Proficient with Unix , Vim , T_EX , Git . Experienced with Docker . Familiar with CI/CD tools.

LANGUAGES

Italian	Native	Spanish	Intermediate proficiency
English	Full professional proficiency	German	Basic proficiency